

Mac, Raspberry Pi, PINE A64 and Docker. The app can also connect to the OpenHAB cloud network, and the IoT devices of the house can be controlled remotely, using a good Internet connection on a smartphone. OpenHAB won the IoT Challenge 2013 and the JavaOne Duke's Choice Award 2013. It was nominated for the JAX Innovation Award 2014 and was the People's Choice Winner at the Postscapes IoT Awards 2014/15.

- **Jeedom:** This is FOSS that can be installed on any Linux system, and it can also be used on Android/iOS using the smartphone app. It has multiple functionalities like the capability to manage scenarios, text/sound interaction with home automation systems, history viewing, graph generation, linking of all equipment/connected objects and the personalisation of the interface. The Jeedom mobile app (for iOS and Android) allows you to control your home automation system. The app automatically connects via a QR code without the need for any configuration. You will find on your mobile the same functionalities that Jeedom (scenarios, connected devices, home automation, plugins, etc) offers on PCs, and you can also customise your app with shortcuts and more.

The main features of Jeedom are:

- It manages the safety of goods and people.
- Automates heating for better comfort and energy savings.
- Views and manages energy usage to anticipate expenses and reduce wastage.
- Communicates by voice, SMS, e-mail or mobile applications.
- Manages all the automatic devices at home — shutters, gates, lights, etc.
- Manages multimedia audio and video devices, and other connected objects.

Jeedom has its own Market, like the App Store or Play Store. It allows you to add home automation features and ensures compatibility with new home automation modules, but also lets you customise its fixed and mobile interfaces.

- **Calaos:** This FOSS can be easily installed and used to transform your home into a smart one. Calaos has a long background in home automation and is built in several different layers. It has low level, as well as uniform and powerful, user interfaces. It knows how to talk to hardware and allows you to automate your home easily. It was first developed by a French company of the same name but when the company was closed in 2013, the entire code was open sourced and released under the GPL. A small community started around the project to continue the development and is growing every day. Calaos can be installed on Linux, Mac, Windows, iOS and Android operating systems. The iOS and Android apps can be used for home automation using a smartphone, remotely. Initially, Calaos supported a special kind of hardware called programmable logic controllers (PLCs) to automate a building or apartment. PLCs can work as controllers to control various home devices like wall switches, fans, air conditioners, water heaters, etc, but have to be wired to the PLC. However, the newer version of Calaos also supports GPIO, Zibase and IoT devices.
- Hardware support:** Calaos supports hardware like Wago PLC, Raspberry Pi, Zodianet's ZiBASE, Cubieboard, Squeezebox, CCTV, etc. The development community is improving the app to support newer hardware arriving in the market. Calaos OS is very easy to install and just by burning a memory card,

it's possible to have a complete working software installation.

Software stack: Calaos comes with the following software stack.

- Calaos Server: The core application.
- Calaos Home: A touchscreen user interface.
- Calaos Webapp: HTML5 Web application.
- Calaos Installer: The configuration tool.
- Calaos iOS: An iOS app.
- Calaos Android: An Android app.
- Calaos-OS: The full Linux OS with everything preconfigured.

Do It Yourself (DIY): With Calaos, one can quickly and easily complete an installation. One can get support from the online community regarding any technical issues. The wiki is full of documentation on how it works, and the Calaos forum can help answer any specific question. One can configure the app as required, and as it's open source, it's even possible to modify it.

Apart from the above three software, there are many more open source software available for home automation but most of them don't support iOS and Android OS on a smartphone. Other OSS that can be used for home automation are Home Assistant, Domoticz, OpenMotics, LinuxMCE, PiDome, MisterHouse and Smarthomatic.

With the help of proper hardware and software, it's possible to make home automation more secure and smarter. Home automation also saves overall power consumption and helps reduce electricity bills. As technology improves and more advanced hardware/software come into the market, home automation will become even more convenient. Finally, some day, a sweet home will also become a smart home for everyone. **END** 

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